

Massive Quarks in the LO Trijet Cross-Section

Light-cone vertices, the modified wave functions, the full $\Phi\Phi^*$,
and the divergence separation redone with $m \neq 0$

Worked directly on `Trijet_LO.pdf` ("Trijet divergences", §1.1–§1.3). Its notation ($\zeta, \eta, \xi; R, \tilde{R}, Y, X; \mathcal{D}; \mathcal{W}$) is kept throughout.
Every structure below is derived from explicit light-cone spinors and checked; the massless limit reproduces your (1.2), (1.10), (1.18) **exactly**. Scripts:
`trijet/checks/`.

Read this first. Your PDF is a "Recap" and does not define $\mathcal{D}, R, \tilde{R}, Y, X$ or $\Phi_{\lambda_1\lambda}^{lirm}$. Rather than guess, I *reverse-engineered* the light-cone vertex factors behind (1.2), (1.10), (1.18) and verified that they reproduce all three expressions identically (§1). Everything after that is derived in *your* conventions. The one thing the massless expressions cannot fix is an overall phase of each vertex; it affects only the massless×mass *interference* terms in §4 (flagged there), and none of the physics results.

0. The answer

(a) The divergence structure is unchanged. The cross-section still has a single simple pole $1/\xi$ as the gluon becomes slow; mass does not regulate it. The separation of §5 is term-by-term identical.

(b) The divergent coefficient changes by exactly one substitution — the massless leading-order transverse-photon factor is replaced by the massive one:

$$[\zeta^2 + \bar{\zeta}^2](R_0 \cdot \bar{R}_0) \longrightarrow [\zeta^2 + \bar{\zeta}^2](R_0 \cdot \bar{R}_0) + m^2,$$

with everything else — the universal weight $16\zeta^2\bar{\zeta}^2$, the three eikonal kernels $\mathcal{K}_{\bar{q}\bar{q}}, \mathcal{K}_{qq}, \mathcal{K}_{q\bar{q}}$, the Wilson-line structures $\mathbb{W}^{(0)}$, and hence the B-JIMWLK/BK kernel — **completely unchanged**.

(c) Bessel argument. All mass dependence of the energy denominators is the single replacement

$$Q^2 \longrightarrow Q^2 + m^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right) \quad (3\text{-body}), \quad Q^2 \longrightarrow Q^2 + \frac{m^2}{z_q(1-z_q)} \quad (2\text{-body}).$$

The reason (b) is so simple: the mass term in the **gluon-emission** vertex is $\mathcal{O}(\xi^2)$, so it drops out of the soft limit entirely; only the mass term in the **photon** vertex survives, and it is precisely the $m^2 K_0 K_0$ piece of the massive $\gamma_T^* \rightarrow q\bar{q}$ wave function. The full $\Phi\Phi^*$ (§4) is much richer — four independent transverse structures instead of one, with m^0, m^2, m^4 terms — but all the new pieces are power-suppressed in ξ .

1. The dictionary: your vertex factors, recovered

Write $\bar{\zeta} \equiv 1 - \zeta$, and recall $\zeta + \eta + \xi = 1$ with ζ, η, ξ the +-momentum fractions of quark, antiquark, gluon. The ansatz is the standard LCPT one: a photon vertex carrying the photon polarization index l and a "momentum" index i , and an emission vertex carrying the gluon polarization index r and a momentum index m ,

$$\mathcal{V}_\gamma^{li}(\lambda_1) = a \delta^{li} + i \lambda_1 \epsilon^{li}, \quad \mathcal{V}_g^{rm}(\lambda_{\text{em}}) = b \delta^{rm} + i \lambda_{\text{em}} c \epsilon^{rm},$$

so that $\Phi_{\lambda_1\lambda}^{lirm} = \mathcal{V}_\gamma^{li} \mathcal{V}_g^{rm}$, with l and r summed between Φ and Φ^* and i, m contracted with R^i, Y^m . Summing over $\lambda_1 = \pm 1$ and over l, r ,

$$\Phi\Phi^* = 2(a_{\text{amp}} a_{\text{conj}} + 1)(b_{\text{amp}} b_{\text{conj}} + c_{\text{amp}} c_{\text{conj}}) \delta^{ij} \delta^{mn} - 2(a_{\text{amp}} + a_{\text{conj}})(c_{\text{amp}} b_{\text{conj}} + b_{\text{amp}} c_{\text{conj}}) \epsilon^{ij} \epsilon^{mn}.$$

With z_q the quark's fraction of the photon momentum and the emitter's parent fraction P :

$$a = 2z_q - 1, \quad b = 2(P - \xi) + \xi = P(2 - w), \quad c = \xi = Pw, \quad w \equiv \frac{\xi}{P},$$

i.e. $\mathcal{V}_g^{rm} = P[(2 - w)\delta^{rm} + i\lambda_{\text{em}}w\epsilon^{rm}]$ — the textbook massless $q \rightarrow qg$ light-cone vertex. Channel by channel:

Channel	z_q	a	P	b	c
antiquark emits (1.1)	ζ	$2\zeta - 1$	$1 - \zeta$	$2\eta + \xi$	$+\xi$
quark emits (1.9)	$1 - \eta$	$1 - 2\eta$	$1 - \eta$	$2\zeta + \xi$	$-\xi$

Verified. Substituting these into the formula above reproduces (1.2), (1.10) and (1.18) identically once $\eta = 1 - \zeta - \xi$ is imposed — all six coefficients ($\delta\delta$ and $\epsilon\epsilon$ in three channels), with no free parameter left. The relative sign of c between the two channels is fixed by (1.18) and is the statement that the emitting line's helicity is $\mp\lambda_1$ for the antiquark/quark.

2. The massive vertices

Both are computed from explicit Lepage–Brodsky light-cone spinors,

$$u(k, \lambda) = \frac{k^+ + \beta m + \boldsymbol{\alpha}_\perp \cdot \mathbf{k}_\perp}{\sqrt{k^+}} \chi_\lambda, \quad v(k, \lambda) = \frac{k^+ - \beta m + \boldsymbol{\alpha}_\perp \cdot \mathbf{k}_\perp}{\sqrt{k^+}} \chi_{-\lambda},$$

checked against $\bar{u}u = 2m$, $\bar{u}\gamma^+u = 2k^+$, $\bar{v}v = -2m$ and $\sum_\lambda u\bar{u} = \text{\textcolor{red}{\textbackslash}slashed}k + m$.

2.1 Photon vertex $\gamma_T^* \rightarrow q\bar{q}$

$$\bar{u}(k_1, \lambda_1) \gamma^l v(k_2, \lambda_2) \propto \underbrace{\delta_{\lambda_2, -\lambda_1} [(2z_q - 1)\delta^{li} + i\lambda_1 \epsilon^{li}] \mathbf{P}^i}_{\text{helicity conserving}} - \underbrace{\delta_{\lambda_2, +\lambda_1} \lambda_1 m (\delta^{l1} - i\lambda_1 \delta^{l2})}_{\text{mass}}$$

The mass term has **no transverse-momentum factor** and requires $\lambda_2 = +\lambda_1$; since $(\delta^{l1} - i\lambda_1 \delta^{l2}) = \sqrt{2} \epsilon_{\lambda_1}^{*l}$ it locks the photon helicity to the quark helicity. Because it lives in a different helicity channel, it does not interfere with the first term at leading order — which is why the LO massive dijet is the familiar $[z_q^2 + (1 - z_q)^2] \epsilon^2 K_1 K_1 + m^2 K_0 K_0$.

2.2 Emission vertex $q \rightarrow qg$ in light-cone gauge

$$\bar{u}(p', \lambda') \text{\textcolor{red}{\textbackslash}slashed}\epsilon_{(r)}^* u(p, \lambda) \propto \frac{P}{\xi} \left\{ \underbrace{\delta_{\lambda'\lambda} [(2P - \xi)\delta^{rm} + i\lambda \xi \epsilon^{rm}] \boldsymbol{\kappa}^m}_{\text{helicity conserving}} - \underbrace{\delta_{\lambda', -\lambda} \lambda m \frac{\xi^2}{P} (\delta^{r1} + i\lambda \delta^{r2})}_{\text{mass}} \right\}$$

The helicity-conserving piece is exactly the b, c of §1 — an independent confirmation of the dictionary. **The mass term carries an explicit ξ^2 .** That single fact is what makes the answer in §0(b) so simple.

2.3 Energy denominators

For an n -parton Fock state of a photon of virtuality Q^2 the light-cone energy denominator is $\sum_i (\mathbf{k}_{i\perp}^2 + m_i^2)/z_i + Q^2$. The transverse part is unchanged by the mass, so all the mass dependence is an additive shift of Q^2 :

$$\underbrace{Q^2 + \frac{m^2}{z_q(1 - z_q)}}_{q\bar{q}}, \quad \underbrace{Q^2 + m^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right)}_{q\bar{q}g} \quad (\text{gluon massless}).$$

Hence every $K_{0,1}(Q\mathcal{D})$ in your expressions becomes $K_{0,1}(\mathcal{E}\mathcal{D})$ with

$$\mathcal{E}^2 = Q^2 + m^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right)$$

Consistency check. As $\xi \rightarrow 0$, $\eta \rightarrow \bar{\zeta}$ and $\mathcal{E}^2 \rightarrow Q^2 + m^2 \left(\frac{1}{\bar{\zeta}} + \frac{1}{\zeta} \right) = Q^2 + \frac{m^2}{\zeta \bar{\zeta}}$, while $\mathcal{D}^2 \rightarrow \zeta \bar{\zeta} \mathbf{r}_{xy}^2$. So $\mathcal{E}^2 \mathcal{D}^2 \rightarrow [\zeta \bar{\zeta} Q^2 + m^2] \mathbf{r}_{xy}^2 = \varepsilon^2 \mathbf{r}_{xy}^2$ — exactly the massive LO dijet Bessel argument, with $\varepsilon^2 = \zeta \bar{\zeta} Q^2 + m^2$. The 3-body and 2-body mass shifts match automatically.

3. The four helicity paths

With a mass term at each of the two vertices, the amplitude has four paths. Writing λ_1 for the quark helicity, λ_p for the intermediate (emitting) line and λ_2 for the final antiquark:

	photon	emission	λ_p	λ_2	order	free indices	multiplies
P1	conserving	conserving	$-\lambda_1$	$-\lambda_1$	m^0	i and m	$R^i Y^m$
P2	mass	conserving	$+\lambda_1$	$+\lambda_1$	m^1	m only	Y^m
P3	conserving	mass	$-\lambda_1$	$+\lambda_1$	m^1	i only	R^i
P4	mass	mass	$+\lambda_1$	$-\lambda_1$	m^2	none	1

Paths with different λ_2 cannot interfere. Hence exactly two interference groups,

$$\{P1, P4\} (\lambda_2 = -\lambda_1), \quad \{P2, P3\} (\lambda_2 = +\lambda_1),$$

and $\Phi\Phi^*$ acquires terms of order m^0 , m^2 and m^4 — but **no odd powers**.

A massless trijet has one transverse structure; a massive one has four.
 $R^i \bar{R}^j Y^m \bar{Y}^n$ (from P1×P1), $Y^m \bar{Y}^n$ (P2×P2), $R^i \bar{R}^j$ (P3×P3), 1 (P4×P4), plus the cross terms.
Each comes with its own mixed-space Bessel structure.

4. The massive $\Phi\Phi^*$

Below, all four transverse vectors are written explicitly (R, Y from the amplitude; $\bar{R}, \bar{Y}$ from the conjugate) since the new structures are not all of the amplitude×conjugate form. Recall $\bar{\zeta} = 1 - \zeta$, $\eta = 1 - \zeta - \xi$.

4.1 Antiquark-emission channel (replaces your 1.2)

m^0 — your (1.2), unchanged:

$$8[\eta^2 + \bar{\zeta}^2][\zeta^2 + \bar{\zeta}^2]\delta^{ij}\delta^{mn} - 8\xi(2\zeta - 1)(2\bar{\zeta} - \xi)\epsilon^{ij}\epsilon^{mn} \quad [\times R^i \bar{R}^j Y^m \bar{Y}^n]$$

m^2 — diagonal (sign-unambiguous):

$$P2 \times P2 : \quad 8m^2[\eta^2 + \bar{\zeta}^2]\delta^{mn} \quad [\times Y^m \bar{Y}^n]$$

$$P3 \times P3 : \quad 8m^2 \frac{\xi^4}{\bar{\zeta}^2} [\zeta^2 + \bar{\zeta}^2]\delta^{ij} \quad [\times R^i \bar{R}^j]$$

m^4 :

$$P4 \times P4 : \quad 8m^4 \frac{\xi^4}{\bar{\zeta}^2} \quad [\times 1]$$

m^2 — massless×mass interference:

$$P1 \times P4 + c.c. : \quad +8m^2 \frac{\xi^2 \zeta \eta}{\bar{\zeta}} [(R \cdot Y) + (\bar{R} \cdot \bar{Y})], \quad P2 \times P3 + c.c. : \quad -8m^2 \frac{\xi^2 \zeta \eta}{\bar{\zeta}} [(\bar{R} \cdot Y) + (R \cdot \bar{Y})]$$

Note the interference terms pair two *amplitude-side* vectors, $(R \cdot Y)$ — a structure absent at $m = 0$. This is because P4 carries no free index, so the amplitude's R and Y contract with each other.

4.2 Quark-emission channel (replaces your 1.10)

By $\zeta \leftrightarrow \eta$, $x \leftrightarrow y$, $R \rightarrow \bar{R}$, $Y \rightarrow X$, parent $P = 1 - \eta = \zeta + \xi$:

$$m^0 : \text{your (1.10)}; \quad P2^2 : 8m^2[\zeta^2 + (1 - \eta)^2]\delta^{mn}; \quad P3^2 : 8m^2\frac{\xi^4}{(1 - \eta)^2}[\eta^2 + (1 - \eta)^2]\delta^{ij}; \quad P4^2 : 8m^4\frac{\xi^4}{(1 - \eta)^2}.$$

4.3 Interference channel (replaces your 1.18)

$$m^0 : \text{your (1.18)}; \quad P2 \times P2 : 8m^2[\xi(1 - \xi) + 2\zeta\eta]\delta^{mn}; \quad P4 \times P4 : 8m^4\frac{\xi^4}{\bar{\zeta}(\zeta + \xi)}.$$

The $P2 \times P2$ coefficient is the same second bracket that appears in your (1.18) — as it must be, since $P2$ differs from $P1$ only at the photon vertex.

One caveat, and it is harmless. The massless expressions (1.2), (1.10), (1.18) fix each vertex only up to an overall sign: $(a, c) \rightarrow (-a, -c)$ leaves them invariant. The mass terms have no such freedom, so the *relative* sign between the m^0 path and the mass paths — i.e. the sign of the $P1 \times P4$ and $P2 \times P3$ interference terms — cannot be determined from your PDF alone. Send me the definition of $\Phi_{\lambda_1 \lambda}^{lirm}$ and I will fix it in one line. It does not affect anything below: **every interference term is $\mathcal{O}(\xi^2)$ and drops out of the divergence (§5).** The diagonal terms $P2 \times P2$, $P3 \times P3$, $P4 \times P4$ are sign-unambiguous.

4.4 Instantaneous terms

The instantaneous quark propagator is $\gamma^+/(2k^+)$, which is mass-independent, so the *vertex* structures (1.3), (1.4), (1.11), (1.12), (1.19), (1.20) are not modified. What changes is (i) the spinors they are contracted with, which adds mass terms of the same four-path type, and (ii) the Bessel argument $Q \rightarrow \mathcal{E}$. Since (1.3)-type terms are $\mathcal{O}(\xi)$ and (1.4)-type are $\mathcal{O}(\xi^2)$ already, their mass corrections are $\mathcal{O}(\xi)$ or smaller and likewise do not touch the divergence.

4.5 Wilson lines

Unchanged. $\mathcal{W}$, S , Q and all of (1.5)–(1.8), (1.13)–(1.16), (1.21)–(1.24) are eikonal: the quark crosses the shockwave at fixed transverse position and picks up a Wilson line, independently of its mass, up to corrections suppressed by $m/(\text{large } k^+)$. Mass enters the trijet only through the wave functions.

5. The divergence separation with $m \neq 0$

5.1 Nothing changes structurally

The three statements of the massless analysis survive verbatim:

- **The only singularity is the $1/\xi$ pole.** Mass does not regulate soft gluon emission: the emission vertex's helicity-conserving piece is mass-independent, so $d^3N \sim \mathcal{G}(0)/\xi$ exactly as before.
- **The transverse integrals stay ultraviolet-finite.** The four-term $\mathcal{W}$ combination still vanishes at $\mathbf{z} \rightarrow \mathbf{y}$ (because $\mathbf{y}' \rightarrow \mathbf{y}$ there), and that argument never referred to the wave function.
- **The collinear region is still a boundary, not a pole** — and is now additionally screened by the mass (see §5.4).

So the separation is the same subtraction:

$$\frac{d^3N}{d^3k_1 d^3k_2 d^3k_3} = \underbrace{\frac{\mathcal{G}_m(0)}{\xi}}_{\text{divergent}} + \underbrace{\frac{\mathcal{G}_m(\xi) - \mathcal{G}_m(0)}{\xi}}_{\text{finite}}, \quad \mathcal{G}_m(\xi) - \mathcal{G}_m(0) = \mathcal{O}(\xi).$$

5.2 The soft limit, path by path

Every path's $\xi \rightarrow 0$ behaviour follows from §4:

Term	order in m	scaling as $\xi \rightarrow 0$	survives?
P1×P1	m^0	ξ^0	yes
P2×P2	m^2	ξ^0	yes
P1×P4, P2×P3 (+c.c.)	m^2	ξ^2	no
P3×P3	m^2	ξ^4	no
P4×P4	m^4	ξ^4	no

Only the two paths whose emission vertex is helicity-conserving survive, because the emission mass term costs ξ^2 . Explicitly, for the three channels:

$$\text{antiquark: } 16\bar{\zeta}^2 \left\{ [\zeta^2 + \bar{\zeta}^2] (R_0 \cdot \bar{R}_0) + m^2 \right\} (Y \cdot \bar{Y}), \quad \text{quark: } 16\zeta^2 \left\{ \dots \right\} (X \cdot \bar{X}), \quad \text{interf.: } 16\zeta\bar{\zeta} \left\{ \dots \right\} (X \cdot \bar{Y}).$$

Multiplying by the prefactors $\zeta^2, \eta^2 \rightarrow \bar{\zeta}^2$ and $-\eta\zeta \rightarrow -\zeta\bar{\zeta}$ of (1.1), (1.9), (1.17), all three collapse onto one universal weight:

$$\mathcal{A}_0^{(m)}(\zeta) = 16\zeta^2\bar{\zeta}^2 \left\{ [\zeta^2 + (1-\zeta)^2] (R_0 \cdot \bar{R}_0) + m^2 \right\}$$

with the interference entering with the same relative minus sign as at $m = 0$.

5.3 The divergent part

$$\left[\frac{d^3N}{d^3k_1 d^3k_2 d^3k_3} \right]_{\text{div}} = \frac{1}{\xi} \mathcal{C} \, 16\zeta^2(1-\zeta)^2 \int_{\Xi} \left\{ [\zeta^2 + (1-\zeta)^2] (R_0 \cdot \bar{R}_0) \mathcal{B}_1 + m^2 \mathcal{B}_0 \right\} \\ \times \left\{ \mathcal{K}_{q\bar{q}} \mathbb{W}_{\bar{q}}^{(0)} + \mathcal{K}_{qq} \mathbb{W}_q^{(0)} - 2\Re[\mathcal{K}_{q\bar{q}} \mathbb{W}_{q\bar{q}}^{(0)}] \right\}$$

with $\mathcal{B}_1 = \frac{K_1(\varepsilon \mathcal{D}_0) K_1(\varepsilon \bar{\mathcal{D}}_0)}{\mathcal{D}_0 \bar{\mathcal{D}}_0}$, $\mathcal{B}_0 = K_0(\varepsilon \mathcal{D}_0) K_0(\varepsilon \bar{\mathcal{D}}_0)$, $\varepsilon^2 = \zeta(1-\zeta)Q^2 + m^2$, and $\mathcal{K}, \mathbb{W}^{(0)}$ **exactly as in the massless case.**

The curly bracket in the first line is nothing but the massive leading-order transverse-photon $q\bar{q}$ wave function squared, $[\zeta^2 + \bar{\zeta}^2] \varepsilon^2 K_1 K_1 \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} + m^2 K_0 K_0$. So the whole effect of the quark mass on the divergence is to replace the LO dijet by its massive

version.

5.4 Consequences

- **B-JIMWLK / BK is untouched.** The kernel is eikonal and mass-independent; the identity $\mathcal{K}_{qq} + \mathcal{K}_{\bar{q}\bar{q}} - \mathcal{K}_{q\bar{q}} - \mathcal{K}_{\bar{q}q} = \mathcal{A} \cdot \bar{\mathcal{A}}$, and its coincidence limit $(\mathbf{x} - \mathbf{y})^2 / [(z - \mathbf{x})^2(z - \mathbf{y})^2]$, hold verbatim. Heavy-quark trijets evolve with the *same* small- x equation; the mass sits entirely in the impact factor.
- **No new divergence, and none removed.** Mass cannot cut off $\int d\xi/\xi$, because soft-gluon emission from a massive quark is still eikonal.
- **The finite part is genuinely richer:** it now contains the three extra transverse structures of §3 and the m^4 term, all of which are $\mathcal{O}(\xi)$ or smaller relative to the pole.
- **Dead-cone screening.** The quasi-collinear region $\mathbf{z} \rightarrow \mathbf{y}$ is softened: the emission mass term is $\mathcal{O}(\xi^2)$ and the Bessel argument $\mathcal{ED}$ never vanishes. This matters only if you subsequently integrate the gluon out (there the mass regulates what the jet definition had to regulate at $m = 0$), not for the fully differential trijet.
- **Two mass scales, not one.** $\mathcal{E}^2 = Q^2 + m^2(1/\zeta + 1/\eta)$ depends on η , i.e. on ξ ; expanding it in ξ generates additional $\mathcal{O}(\xi)$ terms in the finite part that have no massless counterpart.

6. What to change in your notes, concretely

1. In (1.1), (1.9), (1.17): $K_1(Q\mathcal{D}) \rightarrow K_1(\mathcal{ED})$ with $\mathcal{E}^2 = Q^2 + m^2(1/\zeta + 1/\eta)$; likewise in the conjugate.
2. Add to each $\Phi\Phi^*$ the mass terms of §4, remembering that they multiply *different* transverse structures — so (1.1) acquires three further terms in which $R^i Y^m \bar{R}^j \bar{Y}^n / (Y^2 \bar{Y}^2)$ is replaced by $Y^m \bar{Y}^n / (Y^2 \bar{Y}^2)$, $R^i \bar{R}^j$, and 1 respectively.
3. Leave $\mathcal{W}$, S , Q and all of (1.5)–(1.8), (1.13)–(1.16), (1.21)–(1.24) exactly as they are.
4. In the divergence separation, replace $\mathcal{A}_0(\zeta)$ by $\mathcal{A}_0^{(m)}(\zeta)$ of §5.2. Nothing else changes.

6.1 Verification log

Statement	Method	Result
Vertex factors a, b, c reproduce (1.2), (1.10), (1.18)	sympy, on-shell $\zeta + \eta + \xi = 1$	all six coefficients exact
LC spinors: $\bar{u}u = 2m$, $\bar{u}\gamma^+u = 2k^+$, $\sum u\bar{u} = \text{\textcolor{red}{\textbackslash}slashed}k + m$	explicit 4×4	exact
Massive photon vertex structure	explicit spinors	mass term $\propto m$, $\lambda_2 = +\lambda_1$, no $\mathbf{P}$
Massive emission vertex structure	explicit spinors	helicity-conserving part = the b, c of §1; mass term $\propto m\xi^2/P$
LO dijet: $\sum_{\lambda_1, l} \mathcal{V}_\gamma ^2 = 4[\zeta^2 + \bar{\zeta}^2](R \cdot \bar{R}) + 4m^2$	sympy	difference exactly 0
$m \rightarrow 0$ limit of the full massive $\Phi\Phi^*$	sympy	reproduces (1.2) with no extra factor
Soft limits of all three channels share one weight, interference with –	sympy	ratios $+1, +1, -1$ exactly
$\mathcal{E}^2 \mathcal{D}^2 \rightarrow \varepsilon^2 \mathbf{r}_{xy}^2$ as $\xi \rightarrow 0$	algebra	exact